

1 Solution — GIAM 6.6

Problem 6

1.1 Problem

Let $U = \{1, 2, 3, \dots, 10\}$ and $S = \{1, 2, 3\} \subseteq U$. Express the characteristic function χ_S as a set of ordered pairs.

1.2 Setup

The **characteristic function** (or *indicator function*) of a subset S of a universe U is

$$\chi_S : U \longrightarrow \{0, 1\}, \quad \chi_S(x) = \begin{cases} 1 & x \in S, \\ 0 & x \notin S. \end{cases}$$

Any function $f : A \rightarrow B$ can be represented by its **graph** — the set of input-output pairs:

$$\text{graph}(f) = \{(a, f(a)) : a \in A\} \subseteq A \times B.$$

So expressing χ_S as a set of ordered pairs means listing the pairs $(x, \chi_S(x))$ for each $x \in U$.

1.3 Computing χ_S

Apply the rule “1 if in S , 0 if not”:

- $1, 2, 3 \in S$, so $\chi_S(1) = \chi_S(2) = \chi_S(3) = 1$.
- $4, 5, 6, 7, 8, 9, 10 \notin S$, so $\chi_S(x) = 0$ for each of these.

1.4 χ_S as a Set of Ordered Pairs

$$\chi_S = \{(1, 1), (2, 1), (3, 1), (4, 0), (5, 0), (6, 0), (7, 0), (8, 0), (9, 0), (10, 0)\}.$$

This is a 10-element subset of $U \times \{0, 1\}$ — exactly $|U|$ pairs, one per element of the domain.

1.5 Remark — Subsets $\leftrightarrow$ Characteristic Functions

The assignment $S \mapsto \chi_S$ is a *bijection*

$$\mathcal{P}(U) \longleftrightarrow \{0, 1\}^U,$$

between the power set of U and the set of all functions $U \rightarrow \{0, 1\}$. Every subset is captured uniquely by which elements get tagged with 1; conversely, every $\{0, 1\}$ -valued function on U designates a unique subset, namely $\chi^{-1}(1)$.

For $|U| = 10$, both sides have size $2^{10} = 1024$. This is the origin of the notation 2^U for the power set: a subset of U “is” a function $U \rightarrow 2$ (with $2 = \{0, 1\}$), and the set of such functions has cardinality $2^{|U|}$.